

Trajectory-Based Study of Behavioural Dynamics

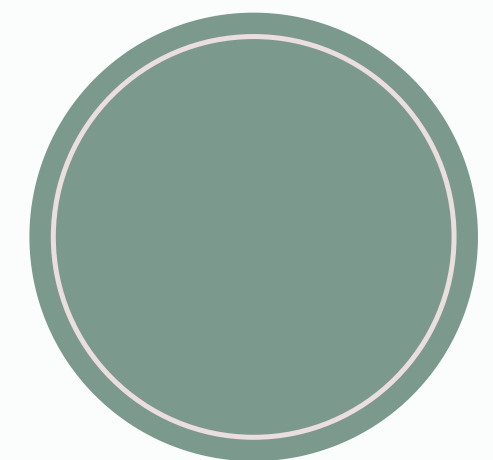
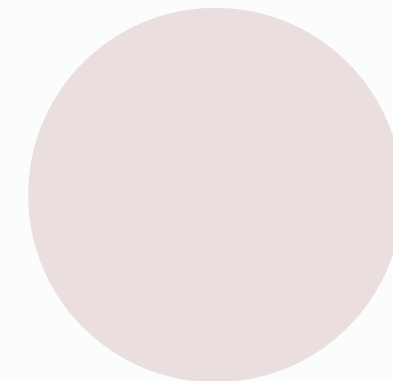
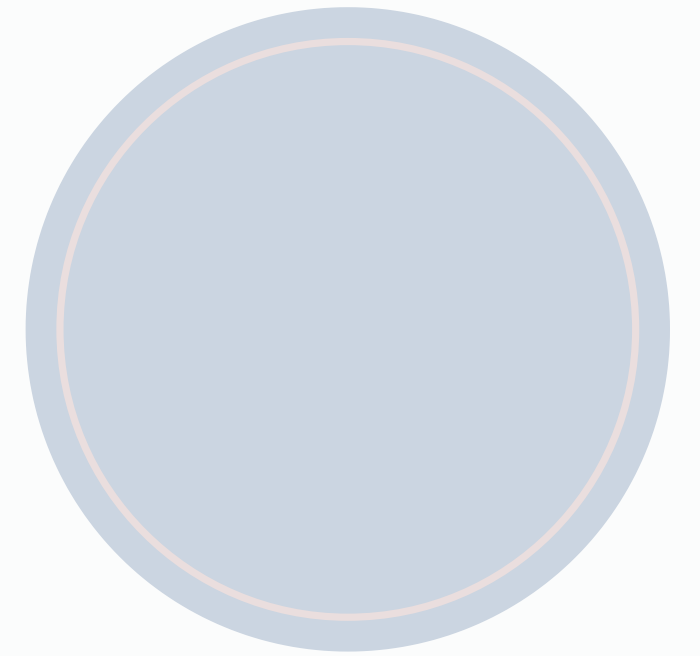
Turbatrix aceti

Under morphine exposure and post-removal conditions

Submitted by
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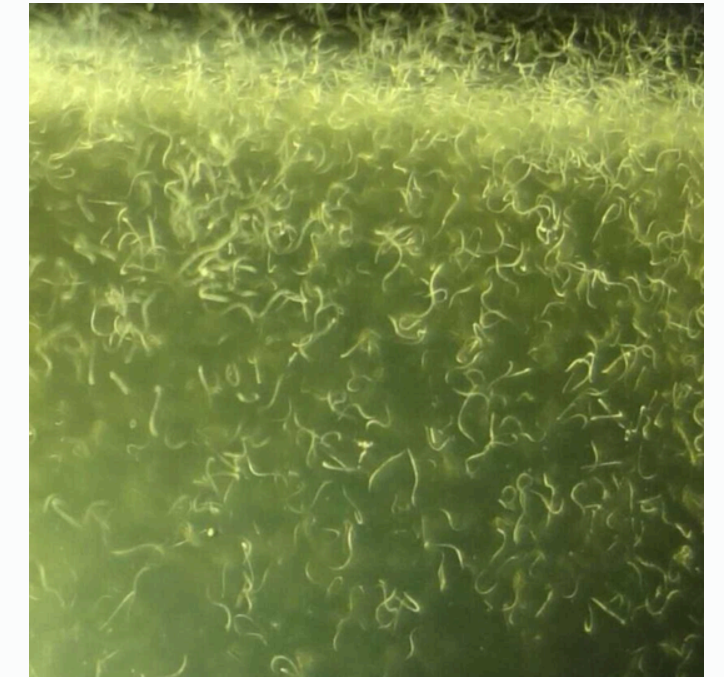
2023CHB1088 · Chemical Engineering

Supervised by Dr. Vishwajeet Mehandia



WHAT IS ACTIVE MATTER?

- Active matter is a system made of particles or organisms that can move on their own.
- They use energy from their surroundings to produce motion.
- Examples include bacteria, algae, and swimming microorganisms.
- Their motion is different from ordinary passive particles, which mainly move due to external forces or random thermal motion.



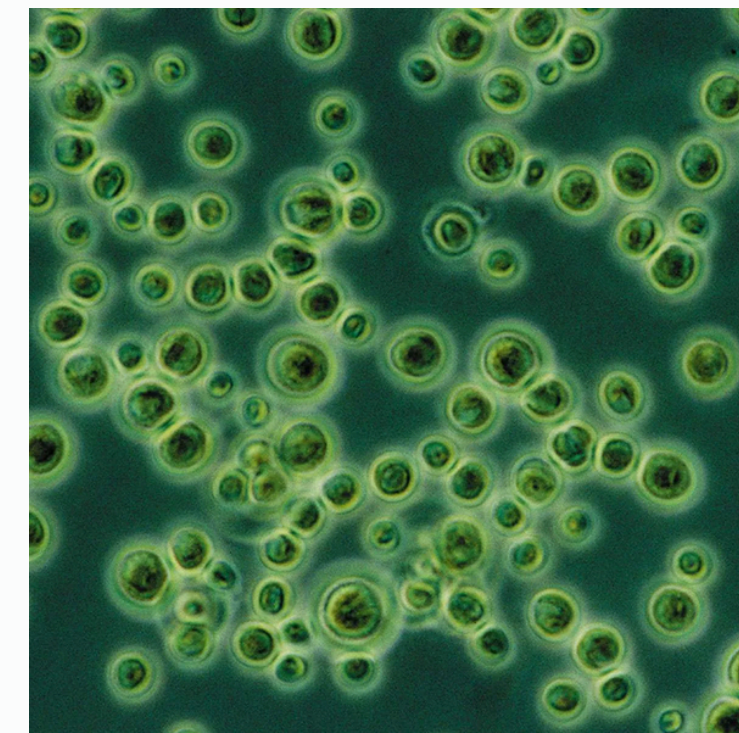
Individual & Collective Behaviour

Individual Behaviour

Each active agent independently responds to its environment and generates motion.

Collective Behaviour

Interactions between many active agents can produce organized patterns and coordinated motion at the group level.



From Swimming Microorganisms to Inertial Active Matter

Active matter

Swimming microorganisms

Self-propelled particle

Inertial active particle

- At microscopic scales, viscous forces generally dominate inertial effects. → **overdamped dynamics**
- This motivates the commonly used overdamped description of active particles.
- However, for larger active particles or low-viscosity environments, inertia can become important. → **underdamped dynamics**

MAIN CONCEPT

Active propulsion + Inertia → new dynamical behavior

How does inertia modify the dynamics of self-propelled particles?

Why Inertia Matters?

Overdamped picture

$$m\dot{\mathbf{v}} \approx 0$$

When inertia is negligible:
The velocity adjusts almost immediately
to the forces acting on the particle

Underdamped picture

$$m\dot{\mathbf{v}} \neq 0$$

When inertia is important:
The particle cannot change its
velocity instantaneously

Underdamped Active Ornstein-Uhlenbeck Model (AOUP MODEL)

Equations of motion

$$\dot{\mathbf{x}} = \mathbf{v},$$

$$\dot{\mathbf{v}} = -\gamma\mathbf{v} + \frac{\mathbf{F}}{m} + \frac{\mathbf{f}^a}{m} + \sqrt{2\gamma\frac{T}{m}}\boldsymbol{\eta},$$

$$\tau\dot{\mathbf{f}}_a = -\mathbf{f}_a + f_0\sqrt{2\tau}\boldsymbol{\xi},$$

Two characteristic timescales

- Inertial relaxation time

$$t_I = \frac{1}{\gamma}$$

$\frac{1}{\gamma} \uparrow \Rightarrow$ velocity relaxes more slowly

- Active persistence time

$$t_P = \tau$$

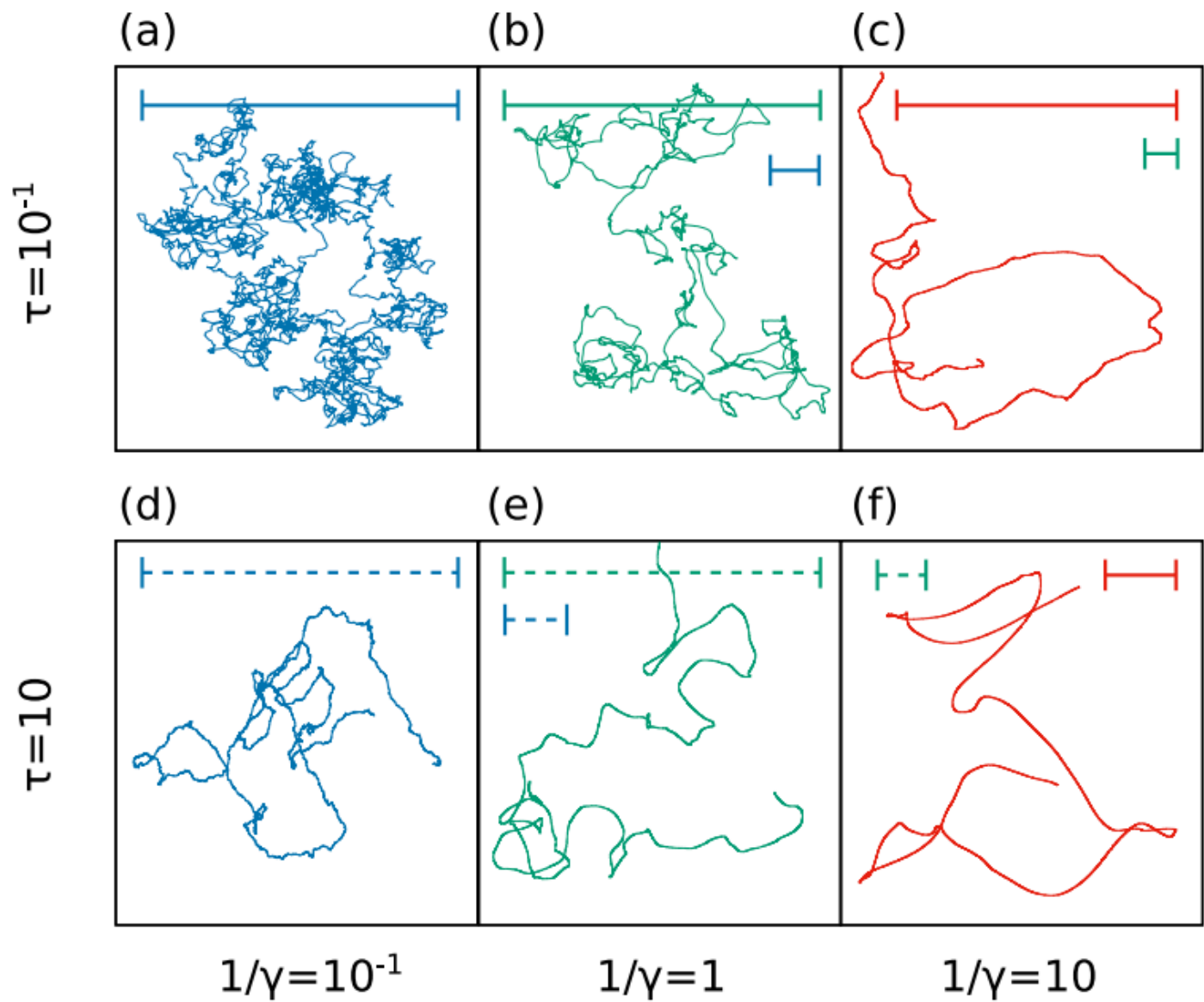
$\tau \uparrow \Rightarrow$ direction changes slowly not frequently

Both these factors makes the trajectory smoother

- Their ratio is

$$\gamma\tau = \frac{\tau}{1/\gamma}$$

which compares the persistence time with the inertial relaxation time.



EXPERIMENTAL WORK

Turbatrix Aceti

commonly known as the vinegar eel — a free-swimming nematode.

- T. aceti is comparatively easy to culture and observe experimentally.
- It has been used before as a model system for studying active biological suspensions — its body size and swimming speed make it convenient for optical observation and trajectory-based measurements.



BODY LENGTH

$\approx 1\text{mm}$

about one millimetre end to end

DIAMETER

$\approx 22\mu\text{m}$

roughly twenty micrometres wide

SWIMMING SPEED

$\approx 0.4\text{mm} \cdot \text{s}^{-1}$

characteristic cruising speed

PROBLEM STATEMENT

The objective of this project is to investigate the behavioural dynamics of *Turbatrix aceti* under different chemical conditions.

Central question

How does morphine exposure affect the locomotory dynamics of *Turbatrix aceti*, and how does the observed motion change after removal of the drug?

Three experimental states considered

Normal / control

Baseline swimming behaviour recorded without morphine exposure.

Morphine exposure

Behaviour observed after exposure to morphine per the experimental protocol.

Post-removal

Subsequent swimming behaviour after the morphine-containing medium is exchanged.

METHODOLOGY

TRAJECTORY BASED STUDY

01

Vinegar-eel suspension

Apple cider vinegar + apple pieces; 10–15 days for a dense population.

02

Apply each condition

Control, morphine exposure, and a post-removal state share the same observation protocol.

03

Image acquisition

AX200 camera at 125 fps, $\approx$ 3 minutes per observation, over 12 days.

04

Trajectory tracking

Fiji / ImageJ + TrackMate workflow converts image sequences into $x,y(t)$ data.

05

Analyse dynamics

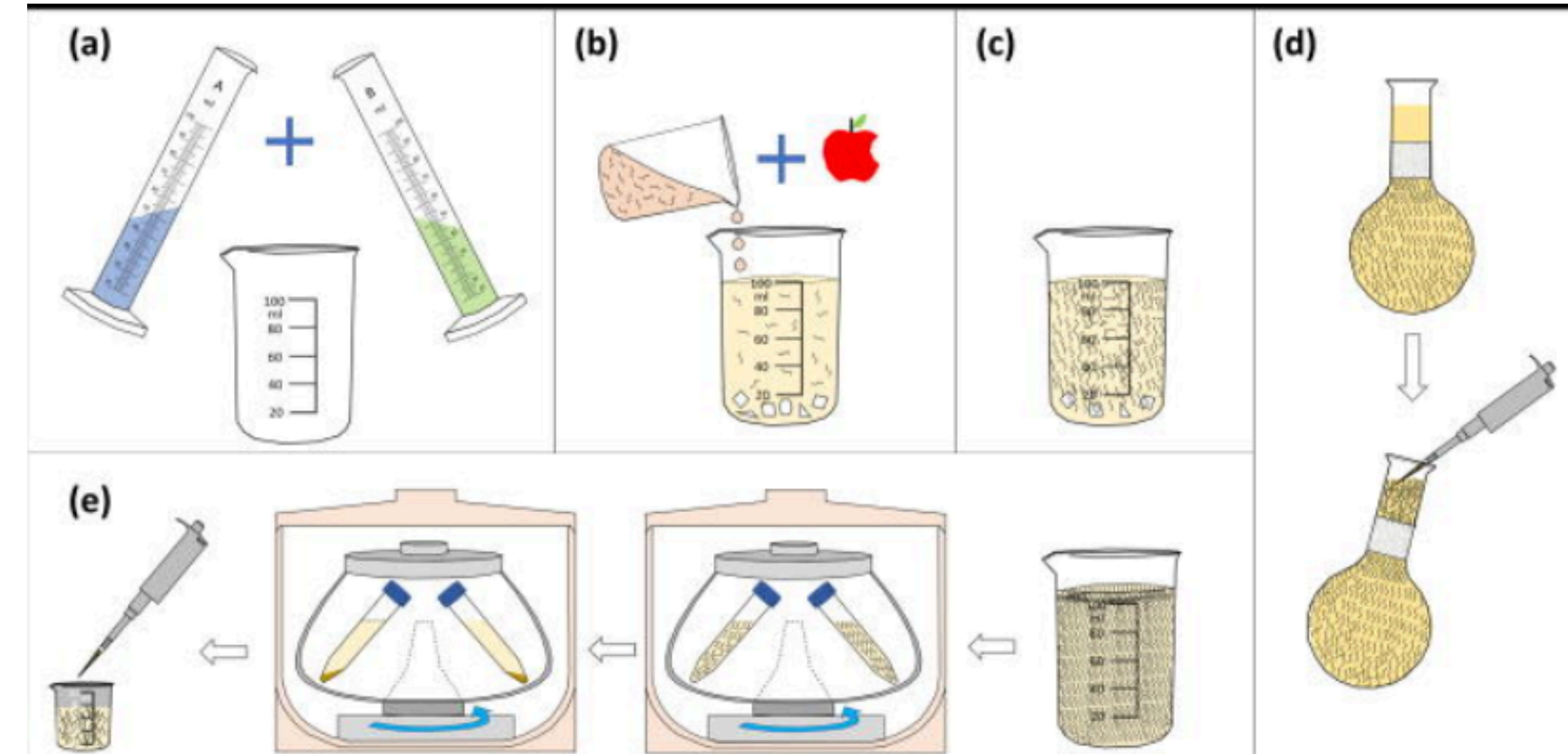
Position, velocity, correlations, **MSD** — compared across the three states.

From *culture* to *trajectories*.

VINEGAR EEL SUSPENSION

INGREDIENTS AND GROWTH

- culture medium — apple cider vinegar with distilled water
- starter culture of *T. aceti* introduced into the medium
- peeled apple pieces added as supplementary nutrient
- covered container with aeration maintained
- $\approx$ 10–15 days for a dense population to develop



IMAGING SETUP

A *bottom-glass* observation chamber.

A bottom-glass Petri dish is used as the observation chamber — the glass bottom provides an optically accessible surface for recording the movement of the organisms. Image sequences are acquired with the AC200 camera at 125 frames per second.

OBSERVATION CHAMBER
Bottom-glass Petri dish

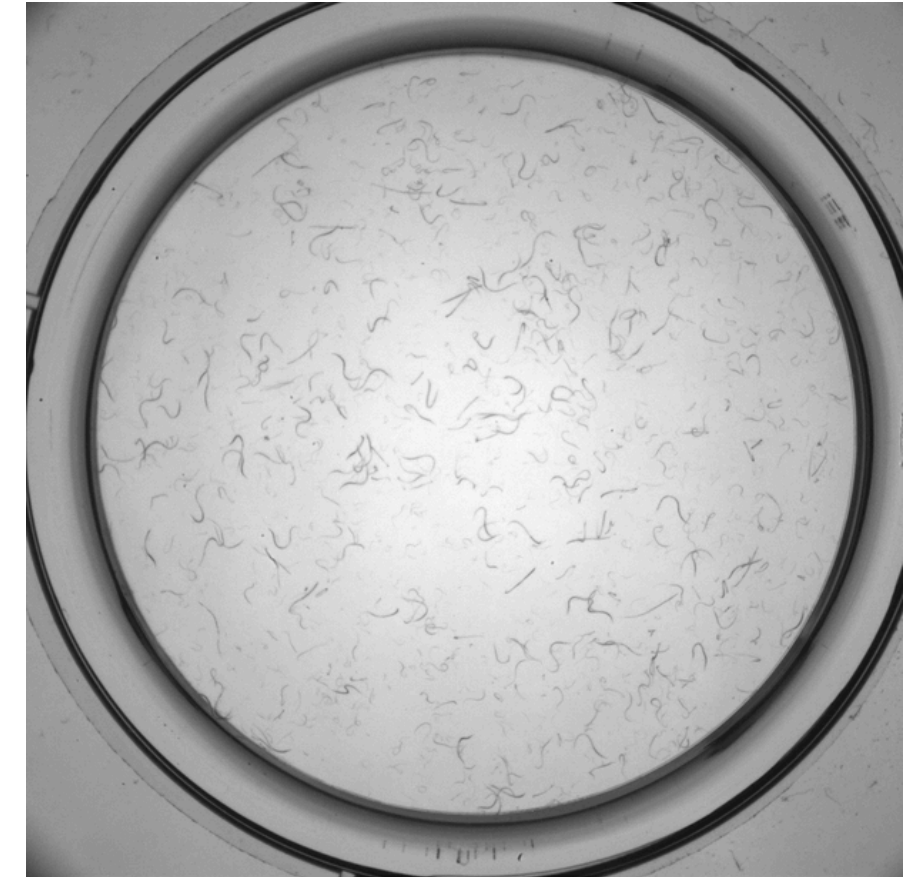
Frame rate
125 fps

OBSERVATION PERIOD
12 DAYS

CAMERA
AX200

RECORDING DURATION
≈ 3 minutes / observation

PRIMARY DATA
Image sequences of
swimming *T. aceti*



SOFTWARE USED & TRACKING

Import into Fiji

The image sequence is loaded into Fiji / ImageJ and converted into a video for visualisation and tracking.

Track individuals

Individual organisms are tracked across successive frames using the TrackMate workflow to obtain a position at each time step.

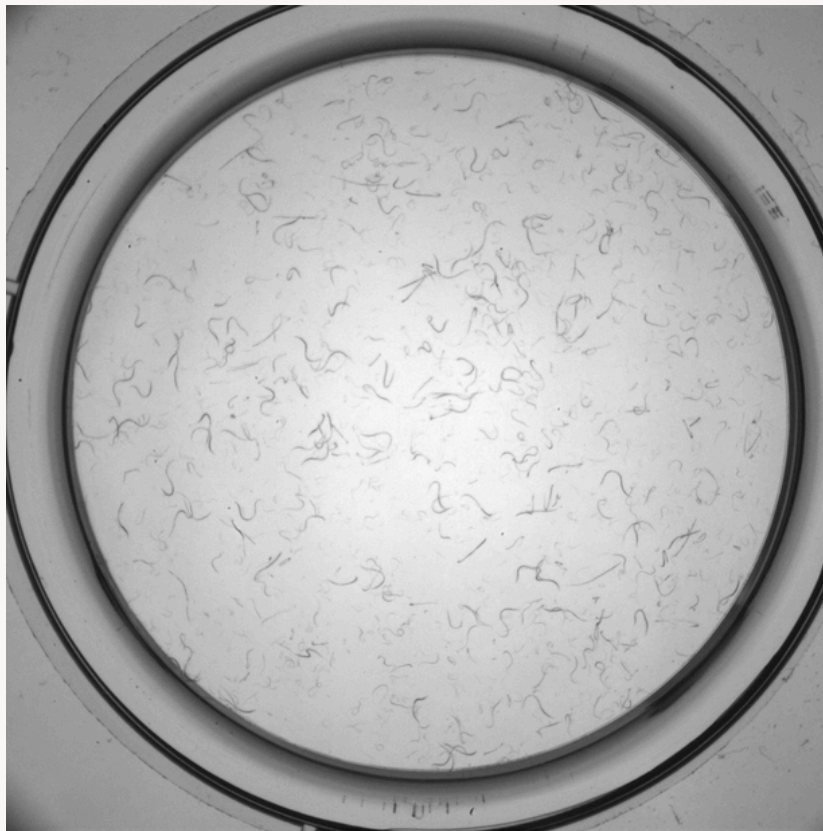
Build trajectories

Successive positions are joined into individual 2D trajectories — the primary dataset for the next stage.

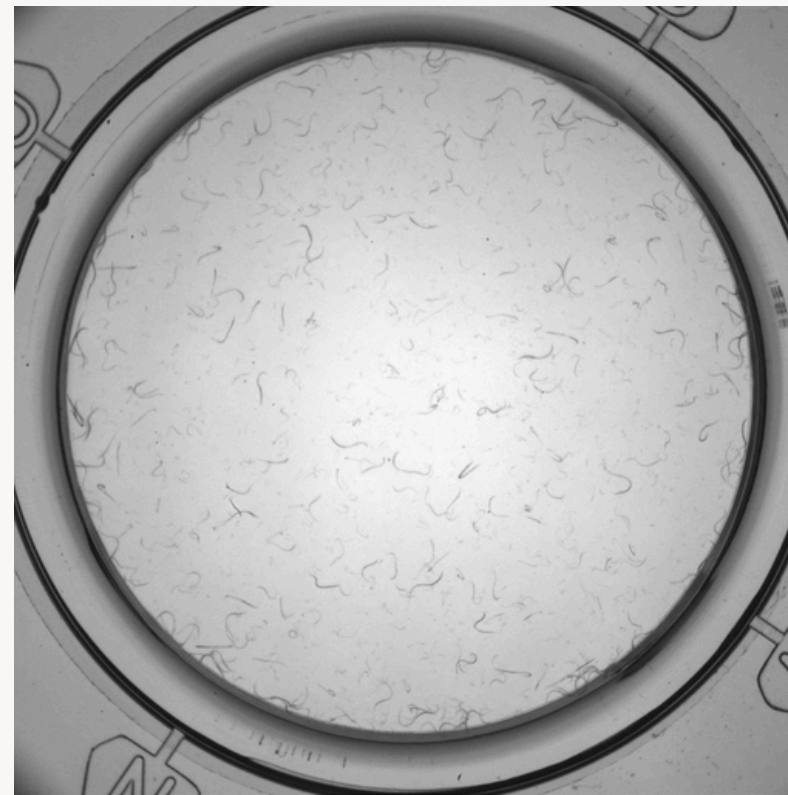
$$\mathbf{r}(\mathbf{t}) = [\mathbf{x}(\mathbf{t}), \mathbf{y}(\mathbf{t})]$$

EXPERIMENTAL OBSERVATIONS

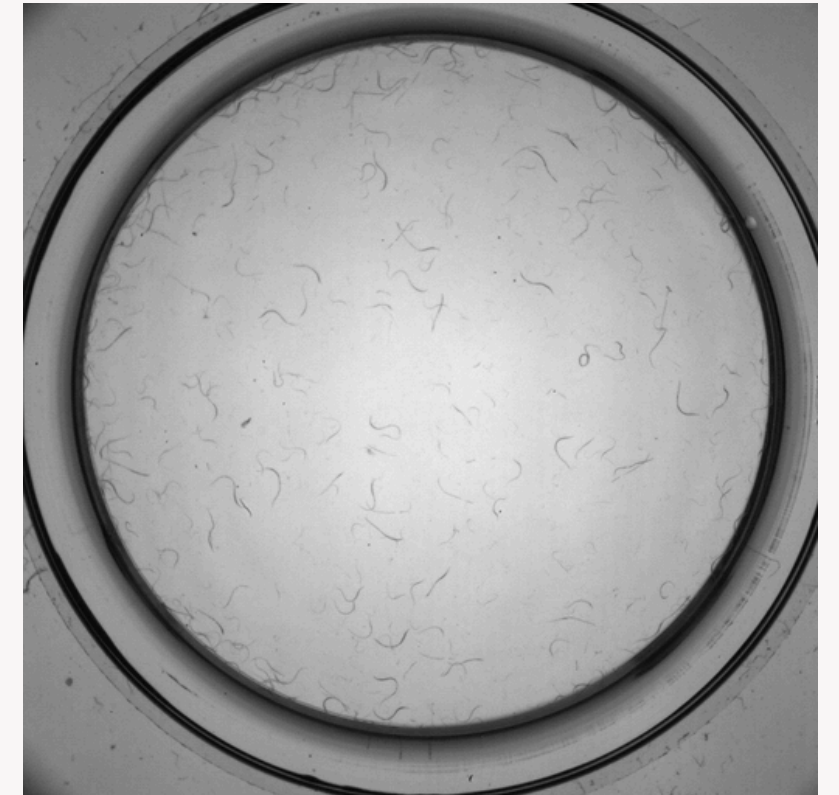
A first look at the *three states*.



**NORMAL/CONTROL
CONDITION**



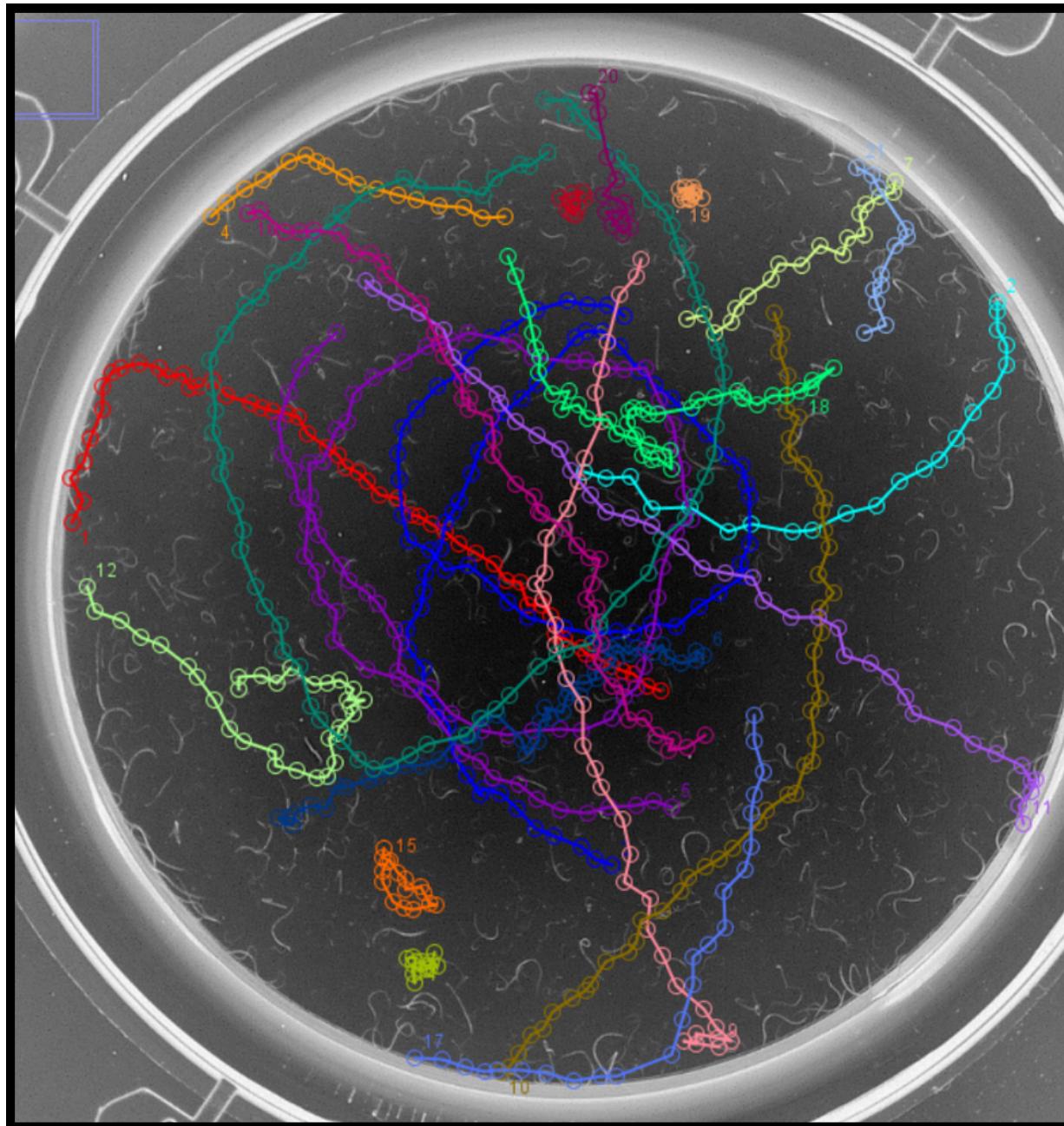
**MORPHINE EXPOSURE
DRUG-TREATED STATE**



**POST-REMOVAL
AFTER MEDIUM EXCHANGE**

TRAJECTORY TRACKING

Many organisms, many paths.



Result

Multiple individuals can be tracked simultaneously within the same observation. Each colour is one organism, with positions joined frame by frame to build individual trajectories.

The resulting paths carry the spatial information needed for quantitative analysis — position, velocity, correlations, MSD.

FUTURE WORK

MEASUREMENTS TO BE CALCULATED FROM THE TRACKED PATHS.

01 · Position

$$\mathbf{r}(\mathbf{t}) = [\mathbf{x}(\mathbf{t}), \mathbf{y}(\mathbf{t})]$$

02 · Velocity

$$\mathbf{v}(\mathbf{t}) \approx [\mathbf{r}(\mathbf{t} + \Delta \mathbf{t}) - \mathbf{r}(\mathbf{t})] / \Delta \mathbf{t}$$

03 · Position correlation

$$\mathbf{C}_r(\boldsymbol{\tau}) = \langle \mathbf{r}(\mathbf{t}) \cdot \mathbf{r}(\mathbf{t} + \boldsymbol{\tau}) \rangle$$

04 · Velocity correlation

$$\mathbf{C}_v(\boldsymbol{\tau}) = \langle \mathbf{v}(\mathbf{t}) \cdot \mathbf{v}(\mathbf{t} + \boldsymbol{\tau}) \rangle$$

05 · Mean square displacement

$$\text{MSD}(\boldsymbol{\tau}) = \langle |\mathbf{r}(\mathbf{t} + \boldsymbol{\tau}) - \mathbf{r}(\mathbf{t})|^2 \rangle$$

What this work establishes, and what comes next?.

Turn trajectories into quantitative measurements, then compare across the three states.

- NORMAL
- DRUG
- POST DRUG REMOVAL

- Position and velocity statistics across conditions.
- Position and velocity correlation functions.
- MSD curves — ballistic transitions, diffusive tails.
- Quantitative answer — does morphine reshape motion, and does it recover on removal?



THANK YOU

